

Midterm assignment Non-Human Cognition 2025

Project proposal

Student number:

S2929619

1. Project title

Comparing the communicative capabilities of indoor and outdoor household cats.

2. General overview of the project, including relevance to science and/or society

The cat is an extremely popular human companion. While in the past cats were semi-domesticated, trading their rodent-hunting labour in for food, nowadays they are a part of many family households all over the world. We have started to care for cats almost as if they were children, feeding them, bathing them, and playing with them. And even though we do not speak the same language, you would be hard-pressed to find a cat owner who has no desire to communicate with their feline. Similarly, cats also frequently give vocal and visual cues when they want our attention.

It is not surprising, then, that research covering communication between cats and humans has been steadily growing bigger. Concerning specifically human-to-cat dialogue, we have come to understand that cats are not only capable of distinguishing between the voices of their owners from strangers (Saito and Shinozuka, 2013), but they can also pick up on whether the speech they hear is directed at them (de Mouzon et al. 2023). Apart from vocal cues, cats have also been shown to comprehend visual signals from humans. Humphrey et al. (2020) discovered that, when humans slowly blink and narrow their eyes at their pets (a positive emotional signal), cats approach more often than when a neutral facial expression is presented. It was even discovered that cats change their behaviour based on if their owner is conveying positive or negative visual and vocal cues (Merola et al., 2015).

However, research regarding the cat's ability to understand human speech with a focus on whether the cat lives an indoor or outdoor lifestyle is lacking. Previous research strongly suggests that the nature of a cat's lifestyle can positively influence their willingness to interact with humans (Turner, 1991) and their tendency to engage in predatory play (Shajid Pyari et al., 2021), with indoor-only cats being more proactive in both aspects compared to cats who have access to the outdoors. Furthermore, both an indoor and outdoor lifestyle come with big benefits and risks for cats, impacting their mental and physical health (News & Views, 2016). If either of these lifestyles prove to have a positive effect on a cat's communicative capabilities, it could be another important factor for cat owners when they decide on their cat's lifestyle.

3. Problem definition, goal and expected results

This proposal argues for a research paper focusing on the difference in communicative capabilities between indoor and outdoor housecats. The frequency with which a cat engages in 'social referencing' will be used to compare this capability. Borrowing the framework used by Merola et al. (2015) during their research on human-to-cat communication, social referencing refers to when a receiver (a cat), looks at an informer (a human) and bases their response to an unfamiliar situation on the informer's positive or

negative emotional reactions. In the same study, Merola discovered that cats do indeed exhibit referential looking when presented with an unfamiliar object, showcasing that they are capable of social referencing. However, the lifestyle of the cat was not considered during the experiment, potentially leaving significant findings uncovered. As mentioned before, the chosen lifestyle for a cat can have a great impact on the cat's willingness to interact with humans (Turner, 1991), meaning that there is value in considering their lifestyle when examining their ability to display social referencing.

The goal of the proposed research will be to examine if there are any differences between indoor and outdoor housecats in their ability to display social referencing. Taking a page out of previous research (Shajid Pyari et al., 2021), this study will refer to indoor housecats as 'indoor-only', and outdoor housecats as 'indoor-outdoor', as cats with access to the outside presumably do not spend an overwhelming majority of their time outside of the house. Consequently, the research question of the proposed study will be: How much does the difference in lifestyle between indoor-only and indoor-outdoor housecats influence their ability to display social referencing?

While multiple rounds of experiments with a substantial number of both indoor-only and indoor-outdoor cats must be conducted to potentially answer the research question, we can hypothesize that, due to the increased willingness to engage with humans, indoor-only housecats might showcase a higher tendency to display social referencing compared to indoor-outdoor housecats. Another insight supporting this hypothesis is that indoor-only housecats have been shown to be more reactive to both visual and auditory stimuli than indoor-outdoor housecats (Shajid Pyari et al., 2021). One could suppose that due to the lack of variety in stimuli that indoor-only housecats receive on a daily basis, they might be more sensitive to novel stimuli in comparison to cats who can freely go outside.

4. Research methodology

I propose a similar methodology used by Merola et al. (2015), as they have already shown a viable way of testing social referencing behaviour in not only cats, but also dogs (Merola et al., 2012). This means that numerous cat-owner pairs should be selected for experimentation, with an approximately equal split of indoor-only and indoor-outdoor housecats. In addition, the cats should also be somewhat used to changing living environments, as the experiment needs to be in an equal, controlled testing environment. We can use the same unfamiliar stimulus introduced by Merola et al. (2015). That being, an electrical fan with plastic green ribbons attached. This seems to induce the desired mild fear reaction that would elicit cats to look at their owner to figure out how to react themselves.

Testing should be done by having the owner and their cat in the testing environment with the electric fan. The owner will then either showcase neutral, positive (curious), or negative (fearful) visual and vocal cues in response to the fan. The behaviour of the cat will then be analyzed to see if the cat will (1) look towards their owner to gauge their reactions, and (2) change their behaviour in accordance with the owner's cues (e.g., approaching the fan if their owner was positive and moving away from the fan if the owner was negative).

5. References

Humphrey, T., Proops, L., Forman, J., Spooner, R. & McComb, K. (2020). The role of cat eye narrowing movements in cat-human communication. *Scientific Reports*, 10, article number 16503.
<https://doi.org/10.1038/s41598-020-73426-0>

Merola, I., Prato-Previde, E. & Marshall-Pescini, S. (2012). Social referencing in dog-owner dyads? *Animal Cognition*, 15, 175-185. <https://doi.org/10.1007/s10071-011-0443-0>

Merola, I., Lazzaroni, M., Marshall-Pescini, S. & Prato-Previde, E. (2015). Social referencing and cat-human communication. *Animal Cognition*, 18, 639-648. <https://doi.org/10.1007/s10071-014-0832-2>

de Mouzon, C., Gonthier, M. & Leboucher, G. (2023). Discrimination of cat-directed speech from human-directed speech in a population of indoor companion cats (*Felis catus*). *Animal Cognition*, 26, 611-619. <https://doi.org/10.1007/s10071-022-01674-w>

News & Views. (2016). AAFP Position Statement: Impact of lifestyle choice on the companion cat: indoor vs outdoor. *Journal of Feline Medicine and Surgery*, 18(9), 752-758. <https://doi.org/10.1177/1098612X16660451>

Saito, A. & Shinozuka, K. (2013). Vocal recognition of owners by domestic cats (*Felis catus*). *Animal Cognition*, 16, 685-690. <https://doi.org/10.1007/s10071-013-0620-4>

Shajid Pyari, M., Uccheddu, S., Lenkei, R. & Pongrácz, P. (2021). Inexperienced but still interested – Indoor-only cats are more inclined for predatory play than cats with outdoor access. *Applied Animal Behaviour Science*, 241, article number 105373. <https://doi.org/10.1016/j.applanim.2021.105373>

Turner, DC. (1991). The ethology of the human-cat relationship. *Schweiz Arch Tierheilkd*, 133(2), 63-70. PMID: 2047832.